

Week 4 Assignment

Task 1: Explore Azure Portal and Create a Resource Group

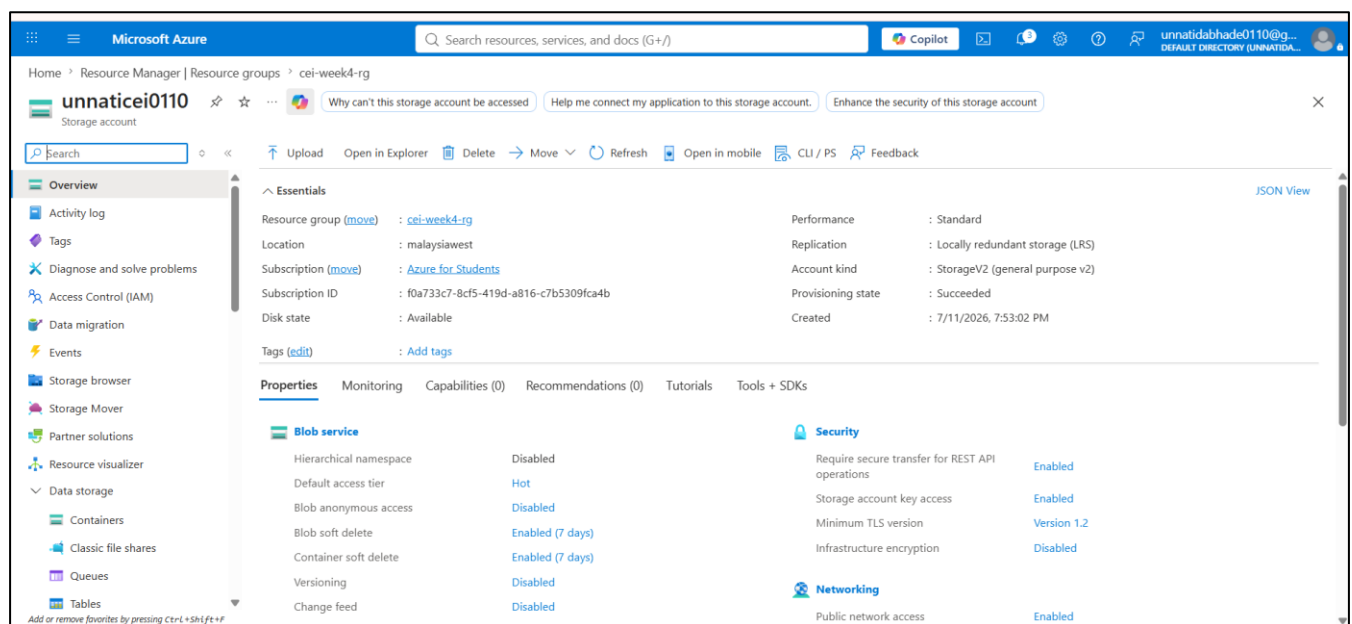
Description

In this task, I explored the Azure Portal and became familiar with its interface. I created a new Resource Group named **cei-week4-rg**, which will be used to organize all the resources required for this assignment.

While creating the Resource Group, I initially tried using the **Central India** region. However, my Azure for Students subscription had a policy that allowed resource deployment only in specific regions. Because of this restriction, Azure did not allow me to create resources in Central India.

After checking the subscription policy, I found that **Malaysia West** was one of the allowed regions. Therefore, I created the Resource Group in **Malaysia West** and continued the assignment successfully.

Screenshot:



Note

Why was Malaysia West selected instead of Central India?

*My Azure for Students subscription has a deployment policy that only allows resources to be created in specific Azure regions. Central India was not included in the list of allowed regions for my subscription. Therefore, I selected **Malaysia West**, which was one of the permitted regions, to complete the assignment successfully.*

Task 2: Storage Setup

Description

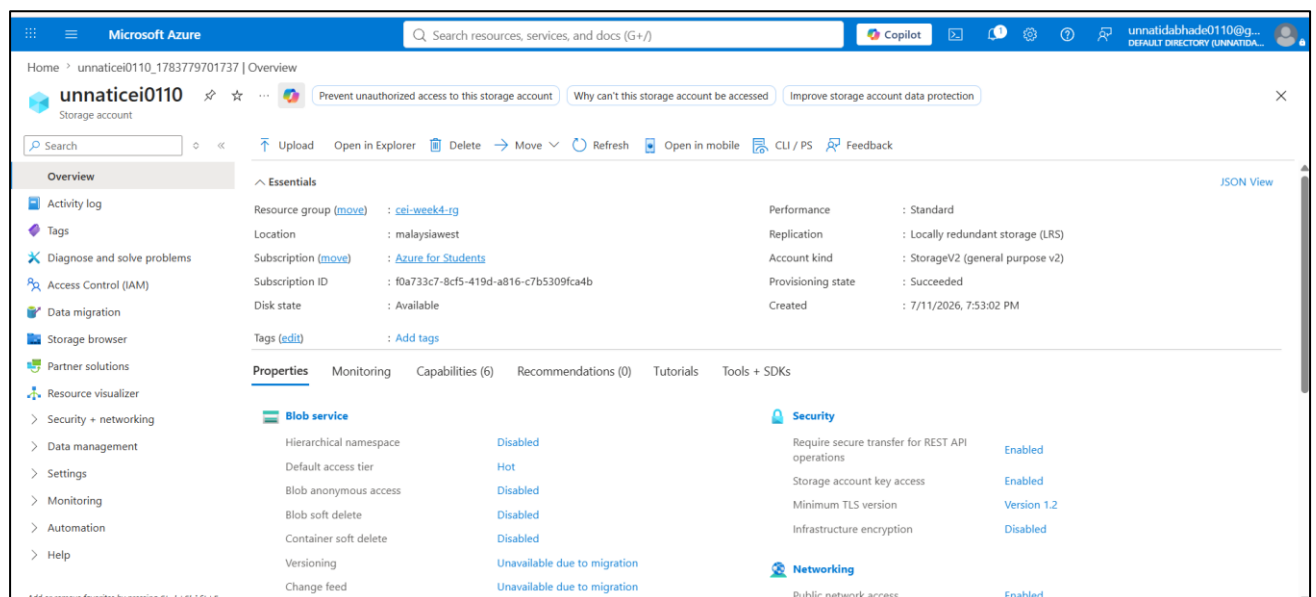
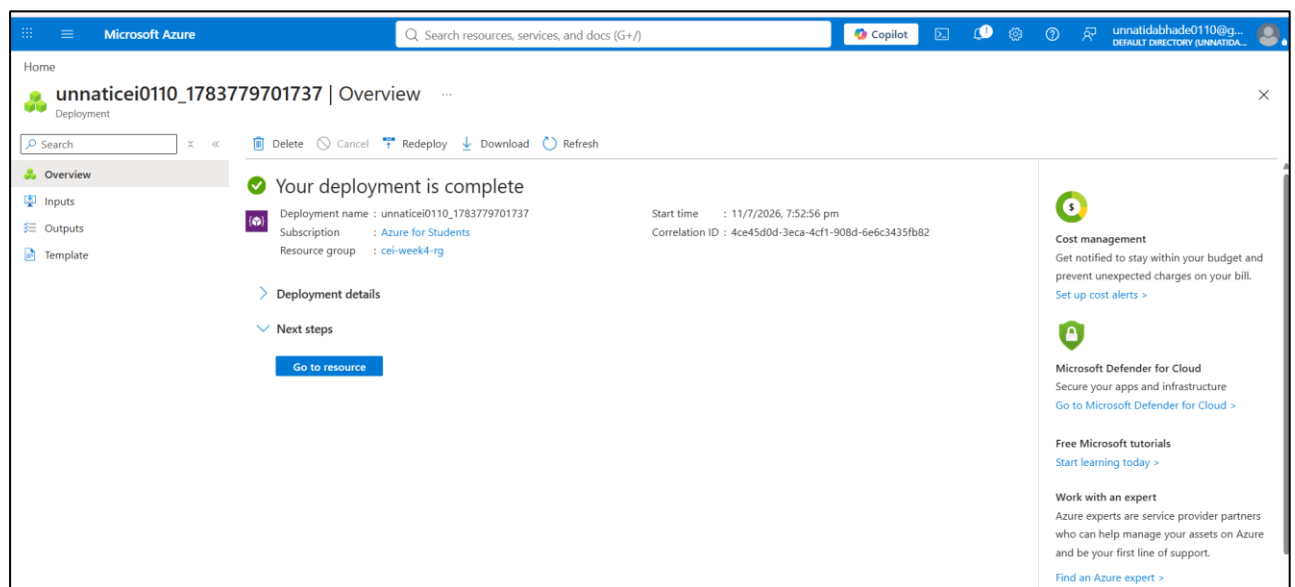
After creating the Resource Group, I created an Azure Storage Account in the same region (**Malaysia West**) to avoid any deployment issues.

Inside the Storage Account, I created a Blob Container and uploaded the **Superstore CSV dataset**, which will be used in the upcoming Azure Data Factory tasks.

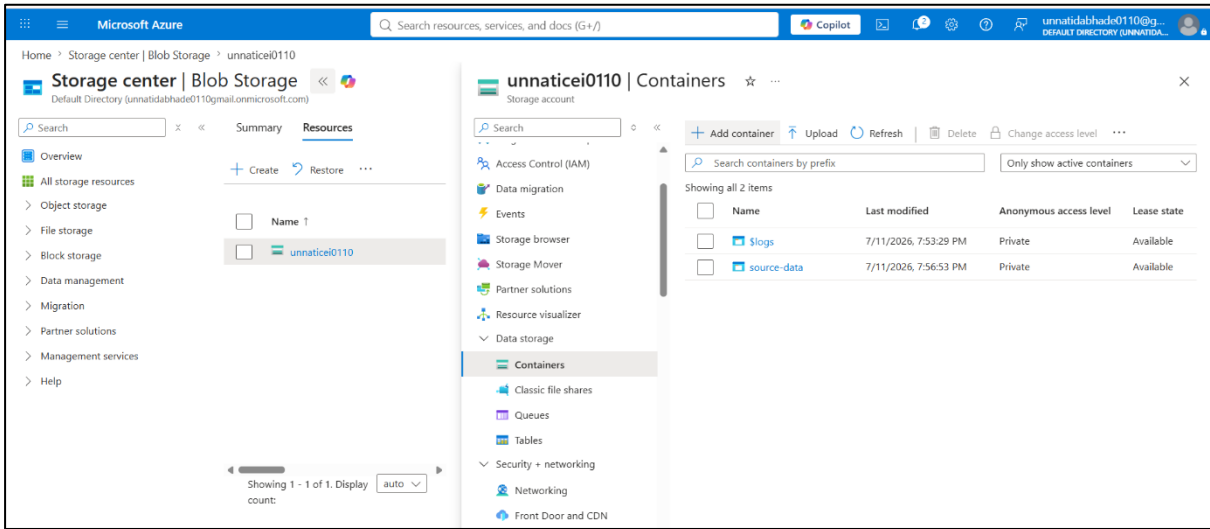
The Storage Account and Blob Container will act as the source storage for building the data pipeline in later tasks.

Screenshot:

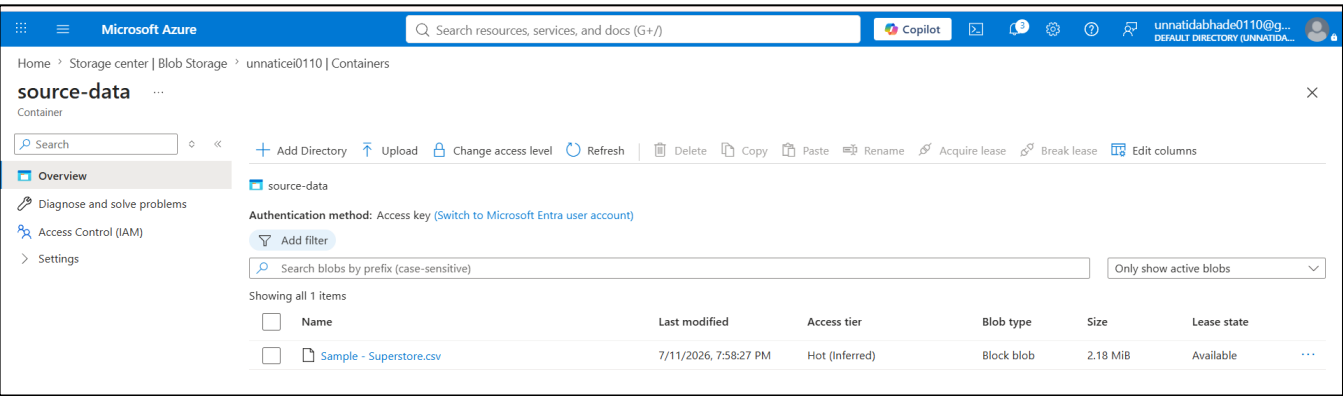
2.1: Creating Storage Account



2.2: Creating a Blob Container



2.3: Uploading a CSV file



Task 3: Azure Data Factory (ADF) Basics

Description

In this task, I created an Azure Data Factory (ADF) instance in the same Resource Group and region that were used in the previous tasks. Azure Data Factory is a cloud-based data integration service that is used to create, schedule, and manage data pipelines.

After creating the Data Factory, I explored the main sections of the ADF Studio, including Author, Monitor, and Manage, to understand their purpose and how they are used while building data pipelines.

Next, I created a Linked Service to establish a connection between Azure Data Factory and the Azure Blob Storage account created earlier. This connection allows ADF to access files stored in the Blob Container.

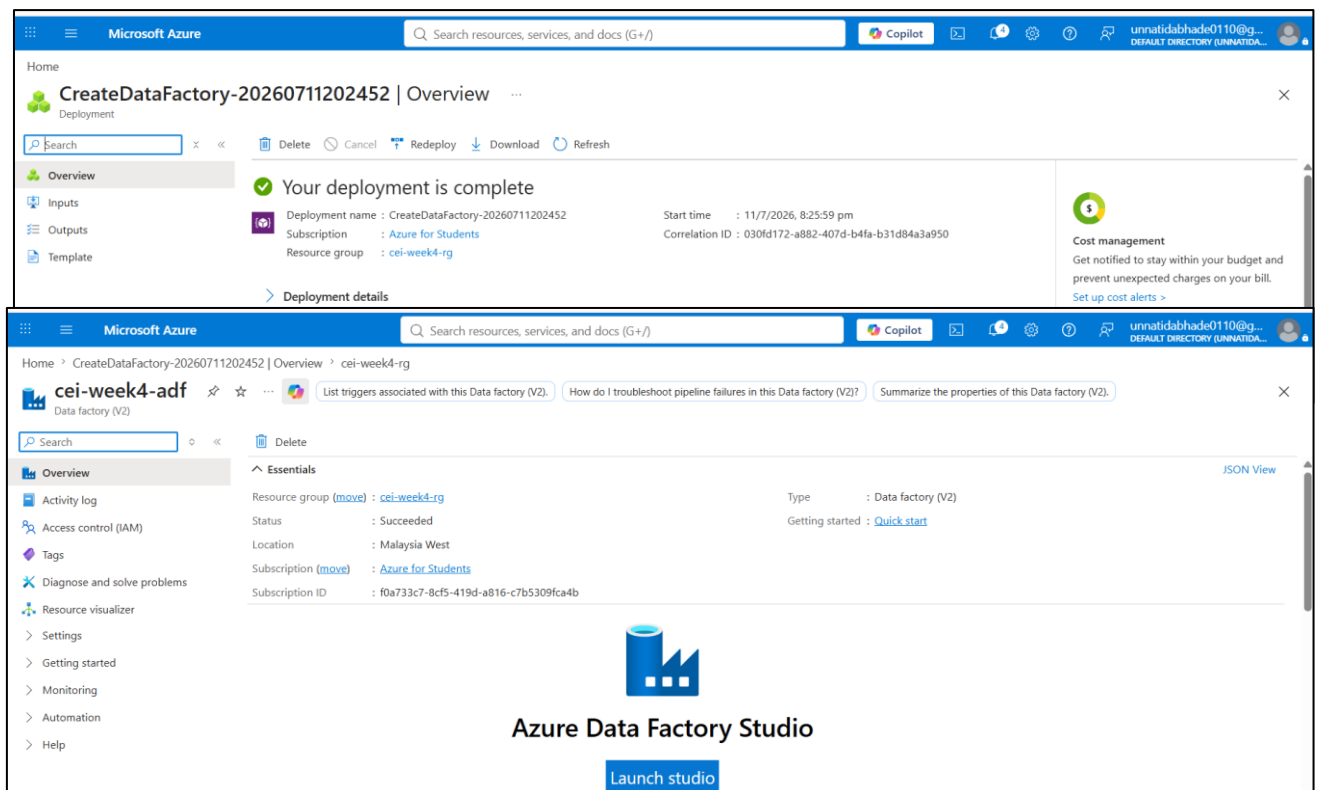
After successfully creating the Linked Service, I created two datasets:

- Source Dataset – connected to the uploaded Superstore CSV file stored in Azure Blob Storage.
- Destination Dataset – configured to store the output file after the pipeline execution.

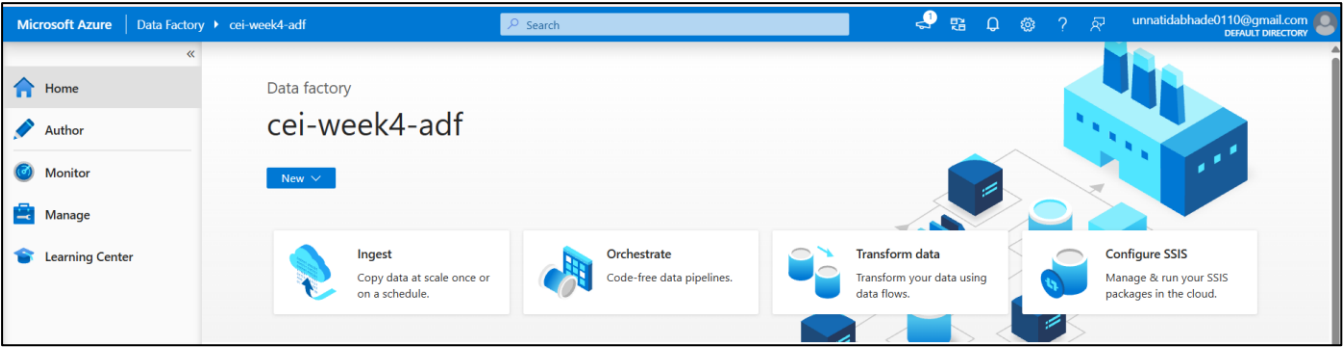
Finally, I created a simple pipeline using the Get Metadata activity. This activity was configured to retrieve metadata such as the file existence, file size, and last modified date from the source dataset. After validating the pipeline, all changes were published successfully.

Screenshots

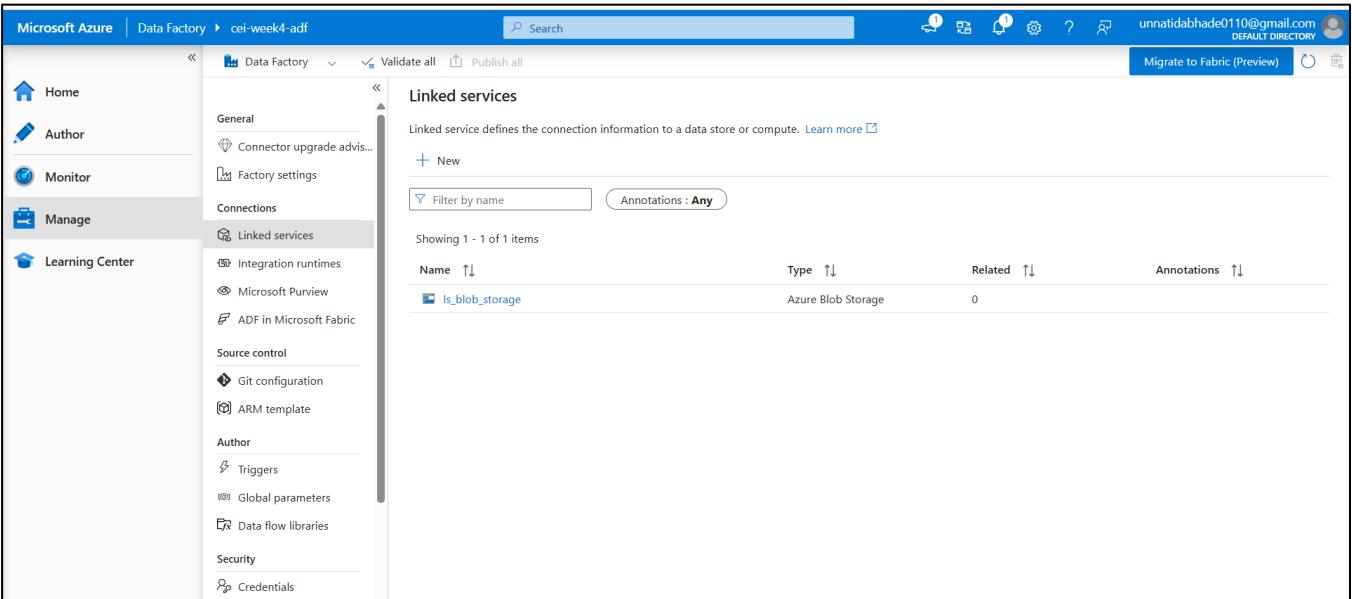
Screenshot 3.1 – Creating Azure Data Factory



Screenshot 3.2 – Exploring ADF UI (Author, Monitor, Manage)



Screenshot 3.3 – Creating Linked Service (Blob Storage)



Screenshot 3.4 – Creating datasets (source + destination)

This screenshot shows the Microsoft Azure Data Factory Author interface. The left sidebar contains navigation options: Home, Author, Monitor, Manage, and Learning Center. The main pane displays the 'Factory Resources' tree on the left, with 'Datasets' expanded to show 'ds_source_superstore'. The central area shows the configuration for this dataset, which is a 'DelimitedText' type. The 'Connection' tab is active, showing the following settings:

- Linked service: **ls_blob_storage**
- File path: **source-data** / **Directory** / **Sam**
- Compression type: **No compression**
- Column delimiter: **Comma (,)**
- Row delimiter: **Default (\n, \r\n, or \r\n)**
- Encoding: **Default(UTF-8)**
- Quote character: **Double quote (")**

The right sidebar shows the 'Properties' pane for the dataset, with the 'General' tab selected. It includes fields for 'Name' (ds_source_superstore), 'Description', and 'Annotations'.

This screenshot shows the Microsoft Azure Data Factory Author interface with the 'ds_destination_superstore' dataset selected. The 'Factory Resources' tree on the left shows two datasets: 'ds_destination_superstore' and 'ds_source_superstore'. The central configuration pane for 'ds_destination_superstore' shows the following settings:

- Linked service: **ls_blob_storage**
- File path: **destination-data** / **output** / **superstore_output.csv**
- Compression type: **No compression**
- Column delimiter: **Comma (,)**
- Row delimiter: **Default (\n, \r\n, or \r\n)**
- Encoding: **Default(UTF-8)**
- Quote character: **Double quote (")**

The right sidebar shows the 'Properties' pane for the dataset, with the 'General' tab selected.

This screenshot shows the 'Preview data' window in Microsoft Azure Data Factory. It displays a table of data from the 'Sample - Superstore.csv' object, linked to the 'ls_blob_storage' service. The table contains 20 columns and 4 rows of data.

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code	Region	Product ID	Category	Sub-Category	Product Name	Sales	Quar
1	1	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gite	Consumer	United States	Henderson	Kentucky	42420	South	FUR-BO-10001798	Furniture	Bookcases	Bush Somerset Collection Bookcase	261.96	2
2	2	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gite	Consumer	United States	Henderson	Kentucky	42420	South	FUR-CH-10000454	Furniture	Chairs	Hon Deluxe Fabric Upholstered Stacking Chairs, Rounded Back	731.94	3
3	3	CA-2016-138688	6/12/2016	6/16/2016	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	California	90036	West	OFF-LA-10000240	Office Supplies	Labels	Self-Adhesive Address Labels for Typewriters by Universal	14.62	2
4	4	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33311	South	FUR-TA-10000577	Furniture	Tables	Bretford CR4500 Series Slim Rectangular Table	957.5775	5

Screenshot 3.5 – Get Metadata Activity

The screenshot displays the Microsoft Azure Data Factory Author interface for a pipeline named 'pl_get_metadata'. The left sidebar shows the navigation menu with 'Author' selected. The 'Factory Resources' pane on the left lists the pipeline and its components. The main canvas shows the 'Get Metadata' activity, which is configured to use the 'Get_File_Metadata' connector. Below the canvas, the 'Output' tab is active, showing the pipeline's status as 'Succeeded' and a table of activity results.

Factory Resources

- Pipelines: 1
 - pl_get_metadata
- Change Data Capture (preview): 0
- Datasets: 2
- Data flows: 0

Get Metadata Activity

Get Metadata (Status: Succeeded)

Get_File_Metadata

Output

Pipeline run ID: 6f66f6b3-759b-4d90-8e03-b08c5aa419eb

Pipeline status: Succeeded

View debug run consumption

Monitor in Azure Metrics | Export to CSV

Showing 1 - 1 of 1 items

Activity name	Activity status	Activity name	Run start	Duration	Integration runtime
Get_File_Metadata	Succeeded	Get Metadata	7/11/2026, 9:08:25 PM	16s	AutoResolveIntegrationRuntime (Malay)

Task 4: Pipeline Development

Description

In this task, I created a new pipeline in Azure Data Factory to copy the data from the source file stored in Azure Blob Storage to a destination location within the same storage account.

A Copy Data activity was added to the pipeline and configured by selecting the source and destination datasets that were created in the previous task. The source dataset was connected to the uploaded Superstore CSV file, while the destination dataset was configured to store the copied file in the output folder.

After configuring both datasets, the pipeline was validated to ensure there were no configuration errors. Once the validation was successful, all the changes were published to Azure Data Factory.

Note

The assignment mentioned using a ForEach activity only if multiple files were available for processing. In this project, only one CSV file was uploaded to the Blob Storage container. Since there was only a single source file, the Copy Data activity alone was sufficient to complete the task. Therefore, the ForEach activity was not required.

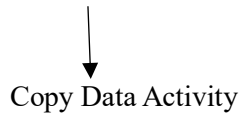
Pipeline Used

Pipeline Name: pl_copy_data

Activity Name: Copy_Superstore_Data

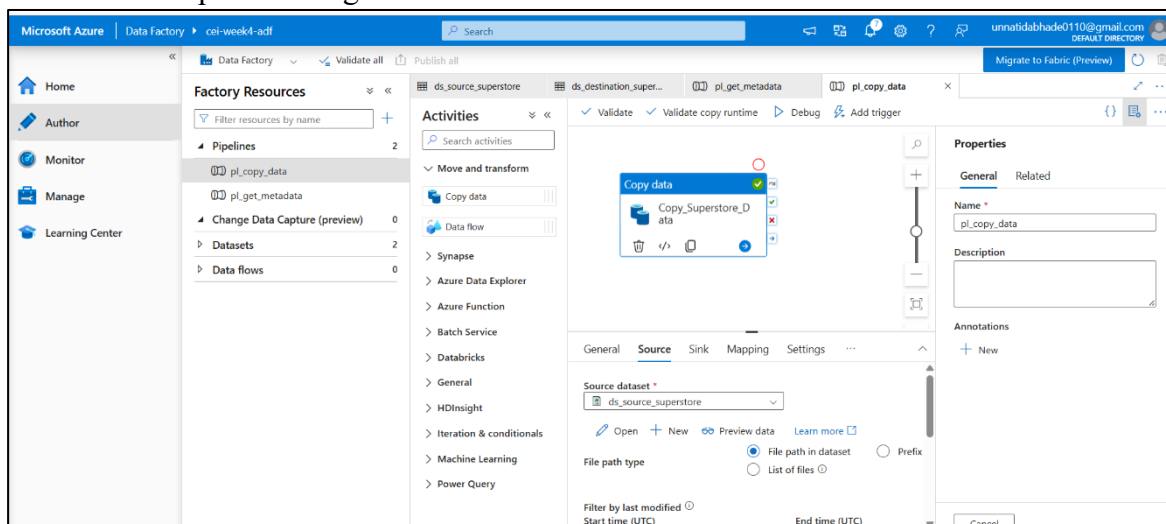
Pipeline Workflow

Source Dataset (Superstore CSV)



Destination Dataset (Output Folder)

Screenshot: Pipeline Design



Task 5: Pipeline Execution

Description

In this task, I executed the data pipeline using the Debug option available in Azure Data Factory. The Debug feature allows the pipeline to be tested immediately without creating a scheduled trigger.

After starting the execution, I monitored the pipeline to verify whether it completed successfully. The pipeline execution finished with the Succeeded status, confirming that the source file was copied successfully from the Blob Storage source to the configured destination.

This step also verified that the pipeline configuration, datasets, and linked service were working correctly.

Pipeline Execution Method

Execution Type: Debug

Pipeline Name: pl_copy_data

Execution Status: Succeeded

Screenshot: Pipeline Execution (Succeeded)

The screenshot displays the Microsoft Azure Data Factory interface. The top navigation bar shows 'Data Factory' and 'cel-week4-adf'. The left sidebar contains navigation options: Home, Author, Monitor, Manage, and Learning Center. The main area is divided into several panels:

- Factory Resources:** A list of resources including Pipelines (pl_copy_data, pl_get_metadata), Change Data Capture (preview), Datasets, and Data flows.
- Activities:** A list of activities including Move and transform (Copy data, Data flow), Synapse, Azure Data Explorer, Azure Function, Batch Service, Databricks, General, HDInsight, Iteration & conditionals, Machine Learning, and Power Query.
- Copy data activity details:** The 'Copy data' activity is selected, showing its configuration and status. The 'Copy data' activity is marked as 'Succeeded' with a green checkmark.
- Properties:** The 'General' tab is active, showing the Name 'pl_copy_data' and a Description field.
- Output:** The 'Output' tab is active, showing the Pipeline ID '9de019fb-a654-4ef8-af-4a-1d61466bc5ba' and the Pipeline status 'Succeeded'. A table below shows the activity status: 'Copy_Superstore_Data' is 'Succeeded'.

Task 6: IAM Roles

Description

In this task, I configured Identity and Access Management (IAM) permissions for the Azure Storage Account.

To allow Azure Data Factory to access the Blob Storage resources, I assigned the Reader and Contributor roles to the Azure Data Factory managed identity.

The Reader role provides permission to view and read the storage resources, while the Contributor role allows Azure Data Factory to perform operations such as creating, modifying, and managing data during pipeline execution.

Assigning these roles ensures that Azure Data Factory has the necessary permissions to read the source data from Blob Storage and write the processed data to the destination without any access issues.

Roles Assigned

- Reader
- Contributor

Assigned To: Azure Data Factory Managed Identity

Screenshot: Role Assignment (Reader and Contributor)

